

Two Touchstones: the Discriminant and the Cubic Formula,

as a Road into Quadratic and Cubic Reciprocity

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Abstract

Two familiar objects, the discriminant of a quadratic and Cardano's formula for a cubic, are used here as fixed touchstones from which quadratic residues, cubic residues, and the arithmetic of the Eisenstein integers $\mathbb{Z}[\omega]$ all grow out of a single question asked at increasing depth: for which primes does a given equation split? The exposition is kept elementary and constructive throughout, with working Python accompanying every stage, culminating in two colour-coded reciprocity matrices, one classical and two-coloured, one cubic and three-coloured, whose contrasting symmetry is itself the punchline.

1 The quadratic touchstone

For $ax^2 + bx + c = 0$ the discriminant is the familiar

$$D = b^2 - 4ac,$$

and its sign over $\mathbb{R}$ does the job everyone learns first: positive gives two real roots, zero a repeated root, negative a complex conjugate pair. The same quantity, however, answers a second and quite different question once we stop working over $\mathbb{R}$ and instead reduce the equation modulo a prime p . There, the equation has a root in $\mathbb{F}_p$ exactly when D is a quadratic residue mod p , or is congruent to 0. This is precisely the statement encoded in the Legendre symbol,

$$\left(\frac{D}{p}\right) = \begin{cases} +1 & D \text{ a nonzero square mod } p, \\ -1 & D \text{ not a square mod } p, \\ 0 & p \mid D, \end{cases}$$

and quadratic reciprocity is the law describing, for a fixed D , exactly which primes p land in each case. It reads, for odd primes $p \neq q$,

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}},$$

so that the two symbols agree unless $p \equiv q \equiv 3 \pmod{4}$, the one case in which they are forced to differ.

A first hand calculation

Take $p = 7$. Squaring the nonzero residues gives

x	1	2	3	4	5	6
$x^2 \bmod 7$	1	4	2	2	4	1

so the nonzero quadratic residues modulo 7 are

$$1, 2, 4.$$

Thus

$$\left(\frac{2}{7}\right) = +1 \quad \text{because} \quad 3^2 \equiv 2 \pmod{7},$$

whereas

$$\left(\frac{3}{7}\right) = -1$$

because no square is congruent to 3 (mod 7). The notation $\left(\frac{a}{p}\right)$ therefore asks a concrete question: does $x^2 \equiv a \pmod{p}$ have a nonzero solution?

For example, the equation

$$x^2 - 3 \equiv 0 \pmod{7}$$

has no solution, while

$$x^2 - 2 \equiv 0 \pmod{7}$$

has the two solutions $x \equiv 3, 4 \pmod{7}$.

Reciprocity in two small examples

Quadratic reciprocity compares two different questions:

$$\left(\frac{p}{q}\right) \quad \text{and} \quad \left(\frac{q}{p}\right).$$

First take $p = 3$ and $q = 7$. Since both primes are congruent to 3 (mod 4), the symbols must have opposite signs. Indeed,

$$\left(\frac{3}{7}\right) = -1, \quad \left(\frac{7}{3}\right) = \left(\frac{1}{3}\right) = +1.$$

Here 3 is not a square modulo 7, although 7, reduced to 1, is a square modulo 3.

For contrast, take $p = 3$ and $q = 5$. They are not both 3 (mod 4), so the symbols agree:

$$\left(\frac{3}{5}\right) = -1, \quad \left(\frac{5}{3}\right) = \left(\frac{2}{3}\right) = -1.$$

The reciprocity formula therefore acts as an exchange law: it tells us what happens when the prime being tested and the prime modulus swap places.

1.1 From polynomial discriminant to field discriminant

Adjoining $\sqrt{D}$ turns the polynomial into a field, $\mathbb{Q}(\sqrt{D})$, and the very same Legendre symbol (D/p) now tells us how a rational prime p behaves there: it splits into two primes when $(D/p) = +1$, stays inert when $(D/p) = -1$, and ramifies when $p \mid D$. Gauss's theory of binary quadratic forms of discriminant D organises the ideal classes of this field into a finite abelian group, the class group, whose order is the class number $h(D)$; the finitely many negative D with $h(D) = 1$ are the Heegner numbers, $D = -3, -4, -7, -8, -11, -19, -43, -67, -163$. We will meet $D = -3$ again shortly, and it will turn out not to be a bystander.

Listing 1: Quadratic residues mod a prime, and the Legendre symbol.

```

1 def legendre(a: int, p: int) -> int:
2     """Legendre symbol (a/p) via Euler's criterion."""
3     a %= p
4     if a == 0:
5         return 0
6     return 1 if pow(a, (p - 1) // 2, p) == 1 else -1
7
8 # quadratic residues mod 13, matching the classical hand-table
9 residues_13 = sorted({(k * k) % 13 for k in range(1, 13)})
10 print(residues_13)           # [1, 3, 4, 9, 10, 12]
11 print(legendre(-1, 13))      # +1 -> 13 = 1 (mod 4), so 13 splits in Z[i]
12 print(legendre(-1, 7))      # -1 -> 7 = 3 (mod 4), so 7 is inert in Z[i]

```

2 The cubic touchstone

Cardano's formula for the depressed cubic

$$t^3 + At + B = 0$$

is

$$t = \sqrt[3]{-\frac{B}{2} + \sqrt{\left(\frac{B}{2}\right)^2 + \left(\frac{A}{3}\right)^3}} + \sqrt[3]{-\frac{B}{2} - \sqrt{\left(\frac{B}{2}\right)^2 + \left(\frac{A}{3}\right)^3}}.$$

The quantity under the square root is a rescaling of the cubic discriminant

$$\Delta = -4A^3 - 27B^2,$$

since

$$\left(\frac{B}{2}\right)^2 + \left(\frac{A}{3}\right)^3 = -\frac{\Delta}{108}.$$

Its sign governs a familiar trichotomy over $\mathbb{R}$. If it is positive, the cubic has one real root and a complex-conjugate pair. If it is zero, the cubic has a repeated root. If it is negative, the cubic has three distinct real roots, although Cardano's expression reaches them through genuinely complex intermediate quantities. This is the *casus irreducibilis*.

In that final case one may instead write

$$t = 2\sqrt{-\frac{A}{3}} \cos \theta,$$

with θ chosen so that

$$\cos(3\theta) = \frac{3B}{2A} \sqrt{-\frac{3}{A}}.$$

The appearance of the triple angle is the first hint that cube roots of unity are about to become unavoidable.

2.1 The discriminant decides the coarse factorisation type

Now reduce

$$f(t) = t^3 + At + B$$

modulo a prime p not dividing Δ . Over $\mathbb{F}_p$, the cubic has one of three factorisation types:

$$1 + 1 + 1, \quad 1 + 2, \quad 3.$$

That is, it splits into three linear factors, factors as one linear factor times an irreducible quadratic, or remains irreducible. That is into three linear pieces, into one linear piece and an irreducible quadratic, or not at all. The answer is governed by the Frobenius element acting on the roots, and remarkably the coarse alternative, cyclic versus transpositional, is settled by nothing more exotic than whether Δ is a quadratic residue mod p :

$$\left(\frac{\Delta}{p}\right) = +1 \implies \text{Frobenius} \in A_3, \quad \left(\frac{\Delta}{p}\right) = -1 \implies \text{Frobenius} \in S_3 \setminus A_3.$$

Quadratic residuosity, in other words, is still the first filter even for a cubic; it is only within the A_3 case that a genuinely new, cubic phenomenon is needed to go any further.

What is a cubic residue?

Modulo 7, the nonzero cubes are particularly easy to see:

x	1	2	3	4	5	6
$x^3 \bmod 7$	1	1	6	1	6	6

so the nonzero cubic residues are only

$$1 \quad \text{and} \quad 6 = -1.$$

For instance,

$$2^3 \equiv 1 \pmod{7}, \quad 3^3 \equiv -1 \pmod{7},$$

whereas 2 and 3 themselves are not cubes modulo 7.

There is an important difference from the quadratic case. The Legendre symbol has two nonzero values, $+1$ and -1 , which say “square” or “nonsquare”. When $p \equiv 1 \pmod{3}$, the nonzero elements instead fall into three cubic classes. To label those three classes naturally we need three values,

$$1, \quad \omega, \quad \omega^2,$$

and hence we are led from the ordinary integers to the Eisenstein integers $\mathbb{Z}[\omega]$.

3 The cubic residue symbol, and why it forces $\mathbb{Z}[\omega]$

Inside the A_3 case we must still distinguish Frobenius of order 1 from order 3, and this requires a cube analogue of the Legendre symbol. Cubing is a proper three-to-one map on $\mathbb{F}_p^\times$ only when $3 \mid p-1$; for such p , the prime splits in the Eisenstein integers $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$, as $p = \pi\bar{\pi}$. Only relative to a prime π of this ring is the cubic residue symbol defined,

$$\alpha^{\frac{N\pi-1}{3}} \equiv \left(\frac{\alpha}{\pi}\right)_3 \pmod{\pi}, \quad \left(\frac{\alpha}{\pi}\right)_3 \in \{1, \omega, \omega^2\},$$

and cubic reciprocity, for primary π, λ (normalised so that $\pi \equiv 2 \pmod{3}$), states the strikingly clean

$$\left(\frac{\pi}{\lambda}\right)_3 = \left(\frac{\lambda}{\pi}\right)_3,$$

with no twist at all: the asymmetry that quadratic reciprocity concentrates into the single case $p \equiv q \equiv 3 \pmod{4}$ simply does not occur here. This is the paper's central structural observation, and §4 makes it visible.

Definition 1 (Primary Eisenstein prime). An element $\pi = a + b\omega \in \mathbb{Z}[\omega]$ with $3 \nmid N\pi$ is primary if $\pi \equiv 2 \pmod{3}$, i.e. $a \equiv 2 \pmod{3}$ and $b \equiv 0 \pmod{3}$. Each of the six associate classes of a prime (there are six units, $\pm 1, \pm\omega, \pm\omega^2$) contains exactly one primary representative.

A cubic symbol computed by hand

The rational prime 7 splits in $\mathbb{Z}[\omega]$, since

$$7 = N(2 + 3\omega) = 2^2 - 2 \cdot 3 + 3^2.$$

The element

$$\pi = 2 + 3\omega$$

is primary. Modulo π ,

$$2 + 3\omega \equiv 0,$$

and therefore

$$\omega \equiv -2 \cdot 3^{-1} \equiv 4 \pmod{7},$$

because $3^{-1} \equiv 5 \pmod{7}$. Consequently

$$\omega^2 \equiv 4^2 \equiv 2 \pmod{7}.$$

For $\alpha = 2$, the definition gives

$$2^{(N\pi-1)/3} = 2^2 \equiv 4 \equiv \omega \pmod{\pi}.$$

Hence

$$\boxed{\left(\frac{2}{2 + 3\omega}\right)_3 = \omega.}$$

Similarly,

$$3^2 \equiv 2 \equiv \omega^2 \pmod{7},$$

so

$$\left(\frac{3}{2 + 3\omega}\right)_3 = \omega^2.$$

The three possible symbol values identify the three cubic classes; they are richer than a simple yes-or-no test for being a cube.

Cubic reciprocity in numbers

Take the primary Eisenstein primes

$$\pi = 2 + 3\omega, \quad N\pi = 7,$$

and

$$\lambda = -1 + 3\omega, \quad N\lambda = 13.$$

Modulo π , we have $\omega \equiv 4 \pmod{7}$, so

$$\lambda = -1 + 3\omega \equiv -1 + 12 \equiv 4 \pmod{7}.$$

Therefore

$$\lambda^{(7-1)/3} \equiv 4^2 \equiv 2 \equiv \omega^2 \pmod{\pi},$$

and hence

$$\left(\frac{\lambda}{\pi}\right)_3 = \omega^2.$$

Modulo λ , we have

$$-1 + 3\omega \equiv 0 \implies \omega \equiv 3^{-1} \equiv 9 \pmod{13}.$$

Thus

$$\pi = 2 + 3\omega \equiv 2 + 27 \equiv 3 \pmod{13},$$

and

$$\pi^{(13-1)/3} \equiv 3^4 \equiv 3 \pmod{13}.$$

But

$$\omega^2 \equiv 9^2 \equiv 3 \pmod{13},$$

so

$$\left(\frac{\pi}{\lambda}\right)_3 = \omega^2.$$

We have therefore seen cubic reciprocity numerically:

$$\boxed{\left(\frac{\lambda}{\pi}\right)_3 = \left(\frac{\pi}{\lambda}\right)_3 = \omega^2.}$$

3.1 Reduction modulo π , made computational

If $\pi = a + b\omega$ divides an element of $\mathbb{Z}[\omega]$, then in the quotient ring $\mathbb{Z}[\omega]/(\pi) \cong \mathbb{F}_p$ we have $a + b\omega \equiv 0$, so

$$\omega \equiv -a b^{-1} \pmod{p} =: r_\pi.$$

This gives an explicit ring homomorphism $\varphi_\pi(x + y\omega) = x + y r_\pi \pmod{p}$, and the cubic residue symbol reduces to ordinary modular exponentiation in $\mathbb{F}_p$, comparing the result against 1, r_π , and r_π^2 .

Listing 2: Finding the primary representative and reducing mod π .

```

1 def primary_rep(a, b, p):
2     """The six unit-associates of a+b*w; return the one with a=2, b=0 (mod
3     3)."""
4     assocs = [(a, b), (-b, a - b), (b - a, -a),
5               (-a, -b), (b, b - a), (a - b, a)]
6     for x, y in assocs:
```

```

6         if x % 3 == 2 and y % 3 == 0:
7             return x, y
8         raise ValueError(f"no primary associate for p={p}")
9
10    def find_ab(p):
11        """ $a^2 - a*b + b^2 = p$  (solvable exactly when  $p = 1 \pmod 3$ , or  $p = 3$ )."""
12
13        lim = int(2 * p**0.5) + 2
14        for a in range(1, lim):
15            for b in range(1, lim):
16                if a * a - a * b + b * b == p:
17                    return a, b
18            raise ValueError(f"no representation for p={p}")
19
20    def cubic_symbol(lam_ab, pi_ab, p):
21        """ $(\text{lambda} / \text{pi})_3$  given primary  $\text{pi}=(a,b)$  of norm  $p$ ,  $\text{lambda}=(a',b')$ ."""
22        a, b = pi_ab
23        r = (-a * pow(b, -1, p)) % p          # phi_pi(omega)
24        a2, b2 = lam_ab
25        phi_lambda = (a2 + b2 * r) % p
26        val = pow(phi_lambda, (p - 1) // 3, p)
27        if val == 1: return 0 # symbol = 1
28        if val == r: return 1 # symbol = omega
29        if val == (r * r) % p: return 2 # symbol = omega^2
30        raise ValueError("unexpected residue")

```

4 Two matrices, side by side

Both matrices below index rows and columns by primes, and colour each cell by the residue symbol relating the row prime to the column prime; the diagonal, where the symbol is undefined, is picked out in gold. The classical matrix (Figure 1) uses two colours and shows visible asymmetry precisely where $p \equiv q \equiv 3 \pmod 4$. The cubic matrix (Figure 2) uses three colours, since cubing is three-to-one rather than two-to-one, and is exactly symmetric, a direct picture of the reciprocity law of §3 having no twist term at all.

Listing 3: Verifying the symmetry claim directly, rather than trusting the picture.

```

1 from sympy import primerange
2
3 primes = [p for p in primerange(2, 100) if p % 3 == 1]
4 data = {}
5 for p in primes:
6     a, b = primary_rep(*find_ab(p), p)
7     data[p] = (a, b)
8
9 symmetric = all(
10     cubic_symbol(data[q], data[p], p) == cubic_symbol(data[p], data[q], q)
11     for p in primes for q in primes if p != q
12 )
13 print("perfectly symmetric:", symmetric) # True

```

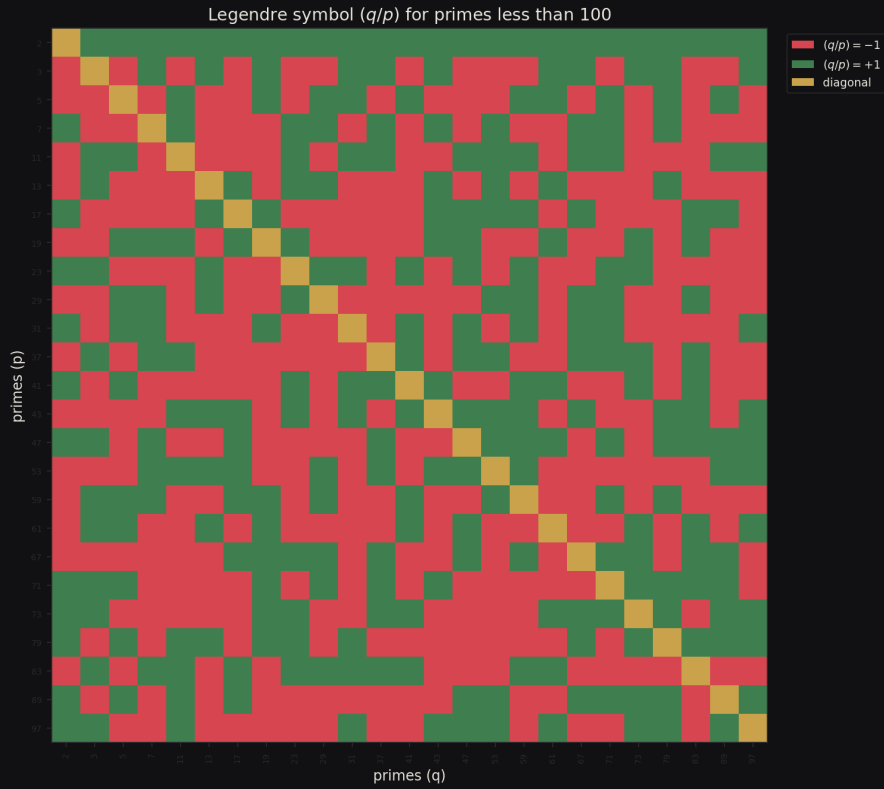


Figure 1: Legendre symbols (q/p) for primes below 100. Colour is determined entirely by $p, q \pmod 4$.

4.1 Why the asymmetry is there in one case and not the other

This contrast is a genuine theorem, not an artefact of how the two matrices happen to have been drawn. Quadratic reciprocity carries an explicit correction term,

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}},$$

which evaluates to -1 exactly when $p \equiv q \equiv 3 \pmod 4$; the two symbols are forced to disagree on that quarter of cells, and Figure 1 shows precisely that quarter. The term is not a bookkeeping inconvenience one could normalise away: $\mathbb{Z}$ has only the two units ± 1 , every prime is already positive, and there is no freedom left to choose a friendlier representative. The sign traces back to the quadratic Gauss sum, $\tau_p^2 = (-1)^{(p-1)/2} p$, and is an intrinsic feature of the real quadratic case.

Cubic reciprocity, once π, λ are both taken primary (Definition 1), has no correction term at all:

$$\left(\frac{\pi}{\lambda}\right)_3 = \left(\frac{\lambda}{\pi}\right)_3.$$

The extra room that makes this possible is exactly the extra room in the unit group: $\mathbb{Z}[\omega]$ has six units rather than two, hence six associates per prime rather than two, and the primary normalisation is precisely the choice of representative, among those six, for which the reciprocity law comes out with no twist. This is a nontrivial fact about $\mathbb{Z}[\omega]$ specifically (small class number, small and tractable unit group), not a general principle that residue laws simplify as the power increases; biquadratic reciprocity in $\mathbb{Z}[i]$, for instance, retains its own correction term even after the analogous primary normalisation. So the red/green asymmetry in Figure 1 and the perfect

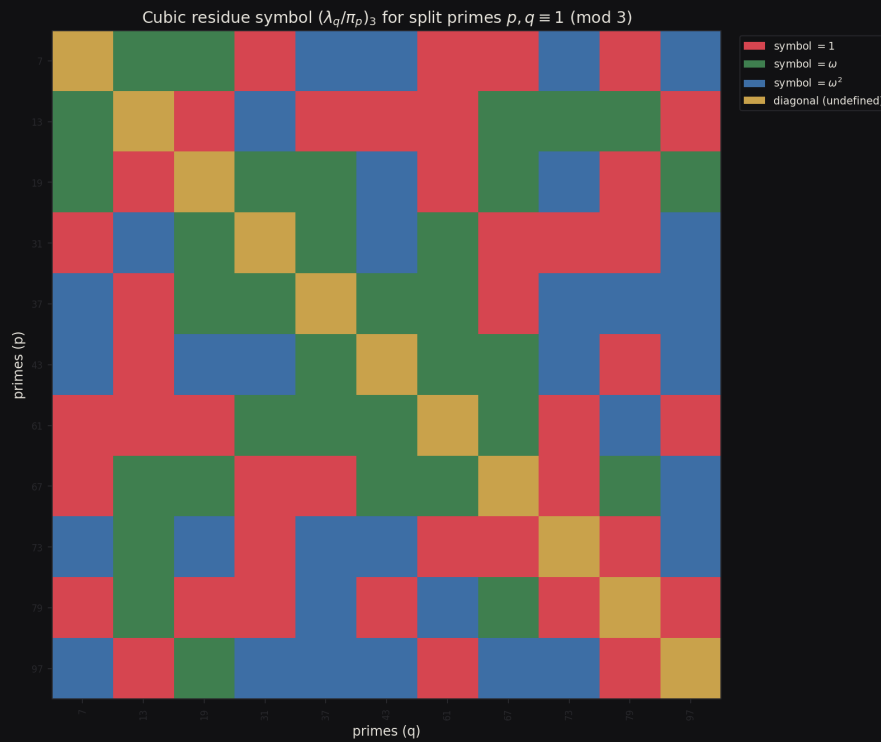


Figure 2: Cubic residue symbols $(\lambda_q/\pi_p)_3$ for primes $p, q \equiv 1 \pmod{3}$ below 100, each represented by its primary Eisenstein prime. The matrix is exactly symmetric.

symmetry in Figure 2 are not the same phenomenon viewed at different resolutions: the first shows a genuine obstruction term built into quadratic reciprocity, and the second shows that, for this particular normalisation, cubic reciprocity simply does not have one.

5 Closing remark: why $D = -3$ was never a bystander

The Heegner number $D = -3$ gives $\mathbb{Q}(\sqrt{-3})$, whose ring of integers is exactly $\mathbb{Z}[\omega]$. It is not a coincidence that this is the field in which cubic residues live: the discriminant $D = -3$ is what you get from asking the quadratic touchstone's question, "adjoin $\sqrt{D}$, what field results," with the answer this time being the one field arithmetically built to host a three-fold, rather than two-fold, notion of residue. The cubic formula's own casus irreducibilis, resolved only by an angle and hence only by the cube roots of unity, was pointing at the same field from the opposite direction all along.